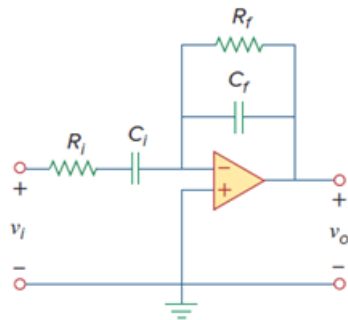


Problem

Obtain the transfer function of the active filter in Fig. 14.91 on the next page. What kind of filter is it?

Figure 14.91:



Step-by-step solution

Step 1 of 1

Refer to Figure 14.91 in the textbook.

Write the impedances of capacitors.

$$\mathbf{Z}_{C_i} = \frac{1}{j\omega C_i}$$

$$\mathbf{Z}_{C_f} = \frac{1}{j\omega C_f}$$

The feedback impedance is parallel combination of $\mathbf{Z}_{C_f}$ and R_f .

$$\begin{aligned} \mathbf{Z}_f &= R_f \parallel \mathbf{Z}_{C_f} \\ &= R_f \parallel \frac{1}{j\omega C_f} \\ &= \frac{R_f \left(\frac{1}{j\omega C_f} \right)}{R_f + \frac{1}{j\omega C_f}} \\ &= \frac{R_f}{1 + j\omega R_f C_f} \end{aligned}$$